

Welcome

Mateus F. V. Pereira

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About Me

I am currently pursuing a Master's degree in Artificial Intelligence at Johns Hopkins University to deepen my knowledge in machine learning, deep learning, and data-driven methods, with a particular interest in exploring the intersection of AI and control systems. My goal is to leverage modern AI techniques, such as reinforcement learning and neural network-based modeling, to push the boundaries of what is possible in advanced control design for complex engineering systems.

I am a control systems engineer with several years of experience in control design applied to aerospace and energy systems. My areas of expertise include constrained control for large-scale nonlinear systems, control of energy conversion systems such as wind turbines, flight control design, and numerical optimization.

I hold a Ph.D. and M.Sc. in Aerospace Engineering from the University of Michigan, where my research focused on developing control architectures for large-scale aerospace systems.

At GE Vernova's Advanced Research Center, I design and deploy advanced control solutions for onshore and offshore wind turbines and other energy systems. I also manage a team building generative AI applications in the energy space.

